

Drone Propulsion Validation - Thrust-to-Throttle Experiment

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Experiment record
Primary Role	System test planner and integration debugger
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Executive Summary

This experiment documented the relationship between throttle command and produced thrust for the drone propulsion system. The goal was to make motor, propeller, ESC, and battery behavior visible before relying on the drone in final flight tests.

The experiment supported safer engineering judgment: instead of choosing parts only from nominal specifications, the team used measured behavior to reason about lift margin, payload feasibility, and control risk.

Applicant Contribution Signals

- Connected propulsion testing to flight stability and payload decisions.
- Used quantitative test behavior to guide final hardware choices.
- Treated the propulsion chain as a system: command input, ESC response, motor output, propeller loading, and battery condition.

Experiment Purpose

Measure how thrust changed with throttle command and identify whether the system had enough margin for controlled flight with the mission payload.

Use the data to support airframe and payload decisions before final integration.

Engineering Interpretation

A useful drone test is not only a number; it is a decision tool. The thrust curve helped identify safe operating regions and exposed where control authority could become fragile.

The result made final testing more deliberate by connecting hardware selection to flight behavior.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Propulsion uncertainty	Measured thrust response	Estimated flight margin before final tests	Test design
Payload requirement	Compared expected lift to mission load	Reduced guesswork in payload integration	Mechanical-electrical integration
Final run safety	Used data to guide throttle and risk planning	Supported more controlled final evaluation	Evidence-based judgment

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.